Building Back, Pushing Out: How Climate Adaptation Drives Displacement in Puerto Rico

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Introduction

The reality of living through a natural disaster is one many people cannot fathom. For Marta Rivera and everyone else on the island of Puerto Rico during Hurricane Maria in 2017, that reality was all too real. Rivera recalls being rushed to the hospital during the storm, only for the building to later collapse while they were still inside (Holpuch et al.). Water poured in, and in the chaos, social security numbers were scribbled on arms with sharpie to identify individuals as they fled the fractured building (Holpuch et al.). Experiences like this are shared collectively by people across the island, as climate change continues to sink its claws into Puerto Rico.

In 2025, climate change is no longer an abstract concern, but a daily reality felt across communities. Nations are well aware of climate change's negative impact on our planet and the consequences are evident ("Causes and Effects of Climate Change"). Fortunately, many nations and individuals are taking proactive measures, implementing new policies and practices aimed at mitigating the harmful effects of climate change and preventing further degradation of our shared environment (United Nations).

While these actions are often praised, we rarely stop to consider the unintended consequences they may bring. Climate adaptation, defined by The Global Center on Adaptation as "taking action to prepare for and adjust to the current and projected impacts of climate change," can itself create new issues for communities facing it (Global Center on Adaptation). This paper focuses on Puerto Rico, an island continuously ravaged by natural disasters and increasingly worried about the effects of climate change. The island experiences a range of climate related weather events, including floods and hurricanes, placing pressure on the island to io adapt to climate change. Homes and neighborhoods are being rebuilt or reinforced to remain

resilient to climate change, but these changes can come at a cost to the communities who have long lived there.

Specifically, I argue that climate adaptation efforts on the island are aiding in displacing the local population, both through forced migration and more subtle forms of cultural and social erosion. Additionally, focusing on the municipality of Rincón provides further evidence of this issue. In this context, the term displacement refers to both the forced migration of individuals from their homes (literal house and neighborhoods), as well as voluntary displacement due to their home no longer representing the culture they once knew.

For more clarity, displacement has been historically caused by a multitude of issues in Puerto Rico. Other arguments as to why Puerto Ricans are being displaced on the island include the influx of Americans moving to PR for tax breaks ("Tax Benefits & Policy - InvestPR"), an ongoing economic crisis (Cheatham and Roy), or the natural disasters faced by the island themselves that are moving people out of their homes (Soto). While all these factors can contribute to displacement in PR, these narratives often overshadow another critical piece: the way the island is being restructured in response to climate change. In some cases, rebuilding efforts prioritize tourism or wealthier populations, leading to gentrification and exclusion of longtime residents.

According to the World Bank's Climate Change Knowledge Portal, Puerto Rico experiences heavy storms and flooding, which have affected hundreds of thousands of individuals on the island (World Bank). Weather events as such are recognized widely as implications of climate change, but what we build, who we build for, and who gets left out deserves equal attention. Rebuilding luxury homes or tourist zones after disasters may strengthen physical infrastructure, but it can also hollow out communities. This topic is extremely

important, as we lack methods to mitigating climate change that consider the lived experiences of people in these locations. As climate adaptation becomes a growing priority, we must ask: how can we address environmental risks without further marginalizing vulnerable populations?

Why Rival Explanations are Missing the Full Story

There is no single explanation for when it comes to displacement, and many times the reason will strongly rely on the circumstances of the situation. One common theory focuses on tax incentives for Americans to move to Puerto Rico, which have resulted in significant debate on the island. Act 60, for example, allows for Americans to move to PR without paying property taxes, a luxury Puerto Ricans themselves are not entitled to ("Tax Benefits & Policy - InvestPR"). These new residents often have the ability to pay for property in cash, making them more attractive to sellers. As a result, properties are increasingly being sold to former mainlanders, raising housing prices and making it harder for locals to afford homes. This results in pushing locals to seek housing elsewhere, even in areas that may not meet their needs because they can no longer afford other areas (Marcos and Mazzei). However, this explanation overlooks how the foundation for luxury living and foreign investment was laid in the first place, often through deliberate development choices and post-disaster rebuilding that cater to wealthier outsiders rather than local residents. Tax breaks may be the vehicle, but the destination is shaped by how the island is being redesigned under the banner of climate resilience.

Another explanation centers on the natural disasters themselves. According to this argument, a community faces a natural disaster that destroys homes, businesses, and other essential community spaces. This argument is logical: in the wake of disaster, people move to safer areas out of necessity. However, it fails to address what happens after a disaster. It assumes

that once the emergency is over residents will return and rebuild, but the reality is far more complicated than that. Many communities are never restored in ways that allow the original residents to come back (Natural disasters cause displacement: UN). The answer often lies in how reconstruction unfolds, who benefits, who is excluded, and how climate adaptation plays into these decisions.

A third explanation points to Puerto Rico's ongoing economic crisis. This crisis has been brought about by multiple factors, including immense debt, government corruption, the Covid-19 pandemic, and natural disasters (Cheatham and Roy). Combined with low median income and an unstable healthcare system, these challenges create large financial pressure on everyday Puerto Ricans, prompting many to leave (Hispanic Federation). While these factors are undeniably real, this explanation doesn't dig deep enough. It fails to ask: what are the solutions being pursued, who are they intended to help, and how do those solutions affect the people already living on the island?

Climate adaptation is the feature missing from these accounts, how it is implemented, and who it is designed to serve. Climate change brings more frequent and intense natural disasters, which leads communities to be rebuilt. But, "such infrastructure building is legitimized as an urgent technical intervention—while less attention is paid to the distribution of costs and benefits among the affected population" (Warner and Weigel). On the island the phrase "reconstruir mejor," or "build back better," has become a guiding slogan, referring to the betterment of affected communities post-disaster (LaFranchi). Post-disaster rebuilding often involves constructing new homes with different materials, relocating communities to less vulnerable areas, and redesigning power systems. These are all critical elements of climate-safe infrastructure, but they're also tied to rising property values, privatization, and gentrification

(LaFranchi). Past research disregards the close connection between climate safe infrastructure and luxury living, missing a key driver of displacement on the island. Understanding how to pursue climate adaptation without displacing local communities is essential as more nations look towards implementing climate safe measures.

Analysis of Outcomes to be Explained

The central outcome explored in this paper is the displacement of local populations in Puerto Rico. As previously mentioned, displacement can take multiple forms: forced migration due to unaffordable housing or destroyed infrastructure, and voluntary migration when a home no longer feels culturally or emotionally recognizable. According to World Bank data, people have been consistently migrating off of the island from the early 2000's to about 2018, when the island experienced its first growth in migrants since 1979. This long-term trend illustrates how deeply rooted and persistent the issue of displacement is on the island.

One of the most direct factors contributing to this displacement is the rising cost of living. In 2023, the average price of a rental unit of all types was $2,463 (Zillow Rental Manager). However, the median household income for Puerto Ricans was $25,096, which is not even enough to pay for 12 months of rent at the average rate ("QuickFacts: Puerto Rico"). This issue is particularly pronounced in coastal regions like San Juan, Rincón, Aguadilla, and Isabela because of their interest to vacationers, with beautiful beaches and lively cultural life. These areas are transforming to accommodate high rises, luxury living, and wealthy tourists, catering more to outsiders than to residents. In Aguadilla and Rincón, international demand for property has skyrocketed housing prices and spurred the construction of luxury developments ("Puerto Rico West Coast: Luxury Living and Investment").

Rincón serves as a strong example of this issue. Located on the northwestern coast, it was once a quiet beach town known for its surf culture and strong community, but it has transformed into a hotspot for luxury tourism. Mariel Muñiz, a self-identified displaced individual from Rincón, described the town as once being a home to hippies who only wanted to enjoy the beach. Muñiz said everyone was respectful of the environment and friendly, and there were no problems with the housing market (González). Today Rincón is known for luxury hotels, high-end Airbnbs, and extravagant mansions. Forty-two percent of its Puerto Rican population now lives in poverty, while vacation rentals can cost over $1,000 per night (González). Additionally, no native generational residents are living in the most popular area of the municipality anymore and many feel unwelcome in spaces that once felt like home (González). Without the ability to afford rising costs, or to relate to the town's changing culture, residents often have no choice but to leave.

Being displaced can result in many issues for the individuals and families affected that are far beyond physical relocation. The first and most immediate impact is economic insecurity. Nearly 3 million Puerto Ricans were classified as impoverished by the U.S. Census in 2023. Figure 1 demonstrates this divide between poverty status for Puerto Ricans and mainland Americans on the island. For people already struggling financially, the additional burden of moving, finding new work, and adapting to unfamiliar communities is immense. Displacement can disrupt employment, education, family responsibilities, and access to healthcare which creates a domino effect that makes long-term stability even more difficult.

In addition to economic insecurity, leaving home causes a loss of cultural identity. A strong connection to place, tradition, and shared values is a cornerstone of identity for many Puerto Ricans. Pride in our homelands and customs, traditions, food, values, and more is evident

in many, but, when people are forced to leave it is very difficult to bring that sense of culture with them for many reasons. One being that without others similar to you in the new area, you may not be able to rekindle the cultural identity. Additionally, and especially when Puerto Ricans move from the island to the mainland, they have to adapt to a completely different cultural identity for purposes of assimilation and safety. When displaced, so much energy has to be placed on creating a safe environment for yourself and your family, which may look like access to food and water, shelter, education, and employment. When those are at the forefront, it is not always easy to maintain one's cultural identity, especially when they might need to let go of it to achieve these necessary survival features first.

One of these cultural barriers is language. Figure 2 demonstrates how 95.8% of Puerto Ricans do not speak English, and many more US born residents report speaking English very well than Puerto Rican born residents. This underscores how language becomes a barrier when English-speaking outsiders move in, and how gentrified spaces often cater to them.

These pressures also take a toll on mental health. The strain of losing one's home, identity, and support systems is emotionally exhausting. Studies show that rates of various mental health disorders including depression, anxiety, and post-traumatic stress disorder are higher among migrant populations, forced or voluntary, than it is the general population (Javanbakht and Grasser). Additionally, the World Health Organization states that these mental health disorders are even higher when the migrant's host population is unwelcoming (World Health Organization). Finally, children are particularly vulnerable because frequent school changes, racism, and instability can hinder their emotional and cognitive development, especially when family cohesion is also disrupted (World Health Organization). Yet, despite the clear

mental health burden of displacement, Puerto Rico continues to suffer from inadequate access to mental healthcare, even before disaster or migration come into play.

Lastly, displacement leads to a breakdown of social capital, the web of relationships, resources, and trust that binds communities together (Mask). When communities face a major hardship, the structure of their relationships can change. The community members people rely on are no longer physically around them and their common resources are gone. This may look like a lack in childcare resources, friendships that promote positive social health, and economic resources such as work. Social capital is critical among poorer communities because what they lack in economic resources they make up for in community engagement, connections, and reliability (Navarra et al.). When people are displaced, these support systems dissolve. Familiar neighbors and family members may no longer be nearby and trusted local institutions like churches, schools, and clinics, are left behind. This social fragmentation weakens community resilience, making it even harder for those affected to recover and thrive.

In short, displacement is not just about leaving a home. It is about losing access to economic stability, cultural roots, emotional well-being, and communal support. While many forces contribute to displacement in Puerto Rico, the island's approach to climate adaptation, focused on "building back better" for outsiders rather than rebuilding for residents, plays a central and under-examined role.

Climate Adaptation as the Problem

Climate adaptation causes displacement in Puerto Rico primarily by altering the socioeconomic landscape of communities. While these efforts aim to mitigate the impacts of climate change through infrastructure improvements like flood barriers and reforestation, they

often result in rising property values and cost of living. This makes it increasingly difficult for local residents to remain in their neighborhoods, while simultaneously attracting wealthier outsiders who can afford the improved conditions. What begins as a protective measure can inadvertently lead to the exclusion of the very people it is meant to help.

The municipality of Rincón illustrates this problem. In 2015, Rincón implemented a community based climate adaptation plan focused on protecting its coastline and addressing the region's growing vulnerability to hurricanes and coastal erosion (Tech). However, the planning process was led solely by municipal officials, designated as stakeholders, without direct involvement from everyday residents. This raises concerns about whether the plan accurately reflected the priorities and experiences of the broader community. These areas were then inspected by Tetra Tech as a technical site visit, and evaluated for potential adaptation efforts. The team focused on seven sites, including a landfill, beaches, wastewater pumping station, and abandoned nuclear reactor.

The findings highlighted critical risks, especially coastal erosion, which threatens not only infrastructure but also the tourism-dependent local economy. The nuclear reactor, a large source of concern, was found to be at risk of leaking due to climate related flooding. Erosion of beaches in Rincón also pose risks to the local economy. The conclusion from all analyses on the selected sites was that there were numerous dangers brought about by climate change, and climate adaptation efforts need to be made to make the community safer and stronger.

The next step of the plan was to hold meetings with community members to discuss more worries regarding climate change. While the plan did eventually include community meetings, these revealed a mismatch between the government’s focus and residents’ concerns. Locals identified critical sites like hospitals and police stations that were initially overlooked. The final

adaptation proposals included beach reforestation, relocating at-risk families, upgrading housing to be hurricane resistant, and improved flood management.

Despite these positive intentions, the implementation of such measures can marginalize local populations. First, by failing to fully incorporate community input from the beginning, the climate adaptation plan neglects important local knowledge. There may also be barriers to participate that Puerto Ricans face, such as lower educational attainment levels. Lower levels of formal educational attainment among many Puerto Rican residents, compared to wealthier transplants, can limit access to jobs in tourism, tech, and real estate sectors, intensifying socioeconomic divides and the risk of displacement (Figure 3). Second, infrastructure improvements often lead to green gentrification. Green gentrification is known as a phenomenon where areas implement more green spaces and climate safe features, which pushes out the existing populations because of raising costs and attracting non-resident investors, tourists, or transplants who price out existing residents (Anguelovski and James JT Connolly). Although the Rincón plan cited goals of strengthening the local economy, the long term consequence may be the transformation of the town into a space designed for outsiders rather than the people who call it home. Finally, the actors driving these efforts, government agencies and private firms, often prioritize their own interests, which may not align with those of vulnerable communities.

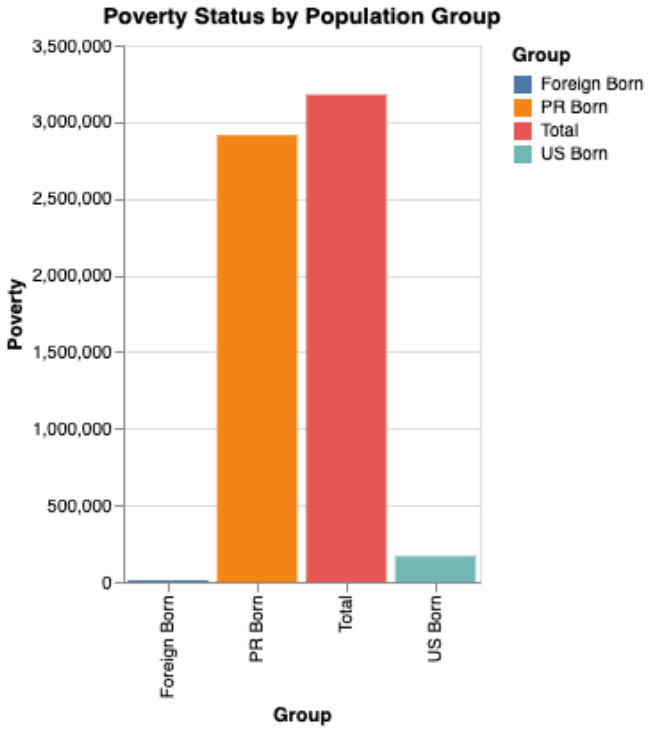
To ensure climate adaptation benefits the people it is meant to protect, future plans must center community voices from the beginning and prioritize justice, equity, and community leadership. First, creating transparent forums for local engagement from the beginning ensures that adaptation efforts are shaped by the lived experiences of local residents. When communities are invited to co-design plans, they are more likely to identify critical sites, voice cultural priorities, and advocate for solutions that reflect their realities (MRSC). This fosters trust and

increases the likelihood that the resulting infrastructure actually serves the people it's meant to protect. Second, investing in cultural preservation such as funding community spaces, protecting historically significant neighborhoods, and supporting local language, arts, and traditions helps to maintain the identity and vibrancy of Puerto Rican communities (EESI). When development actively honors and protects cultural heritage, it strengthens social cohesion and reduces the sense of alienation that often accompanies gentrification and displacement. Third, anti-displacement policies are crucial to make adaptation equitable (Dorazio). Tools such as rent control can keep housing costs stable as neighborhoods improve. Housing stipends or subsidies also can help families facing financial instability stay rooted in their communities rather than being forced to move. Finally, community land trusts, where land is held collectively by a nonprofit or community group, can ensure long-term affordability and prevent speculative real estate development (Georgetown Climate Center). These models shift the focus from private profit to public good.

Together, these measures offer a blueprint for climate resilience that is socially sustainable. They allow Puerto Rican communities to adapt not only to environmental threats, but also to the economic and social pressures that often come with adaptation efforts. In doing so, they preserve the right of residents to remain in place and stay connected to their homes, neighborhoods, and heritage. True resilience lies not just in withstanding the effects of climate change, but in ensuring that the people who live through it are still there to rebuild, thrive, and carry on their ways of life.

Appendix

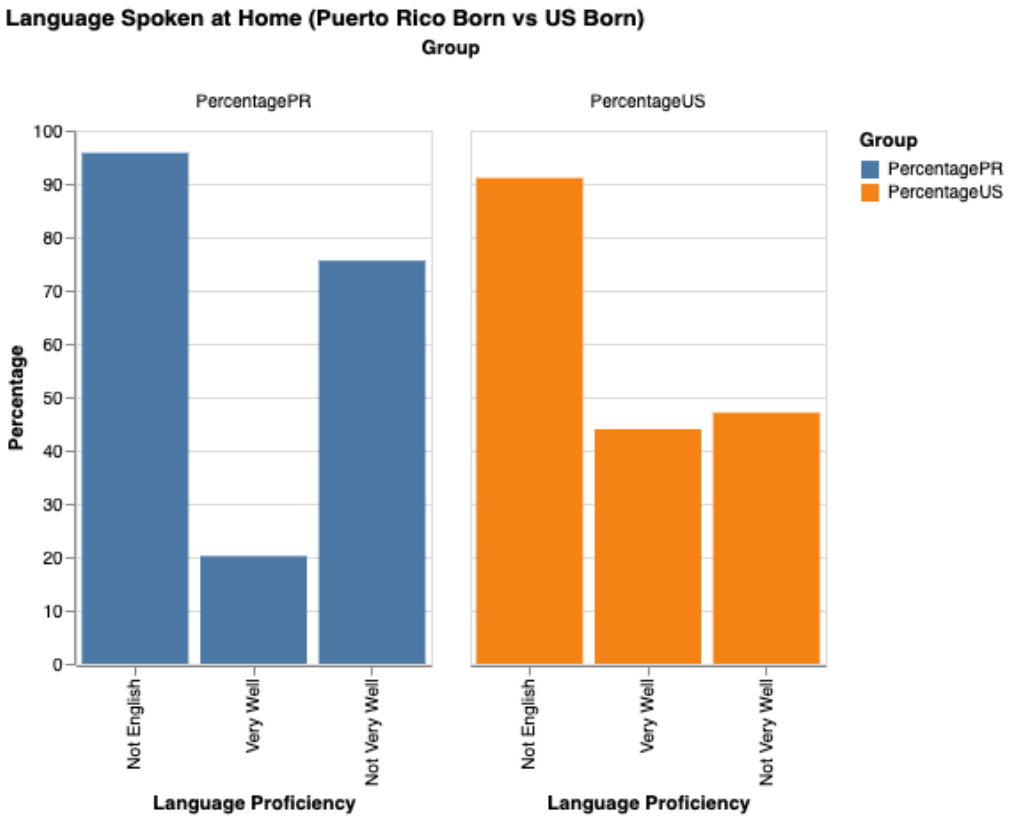
Figure 1: Poverty Status by Population Group



This figure demonstrates the poverty status by population group. It highlights the stark differences in the number of Puerto Ricans that are in poverty compared to the number of mainland Americans that are in poverty on the island of Puerto Rico.

Source: United States Census Bureau

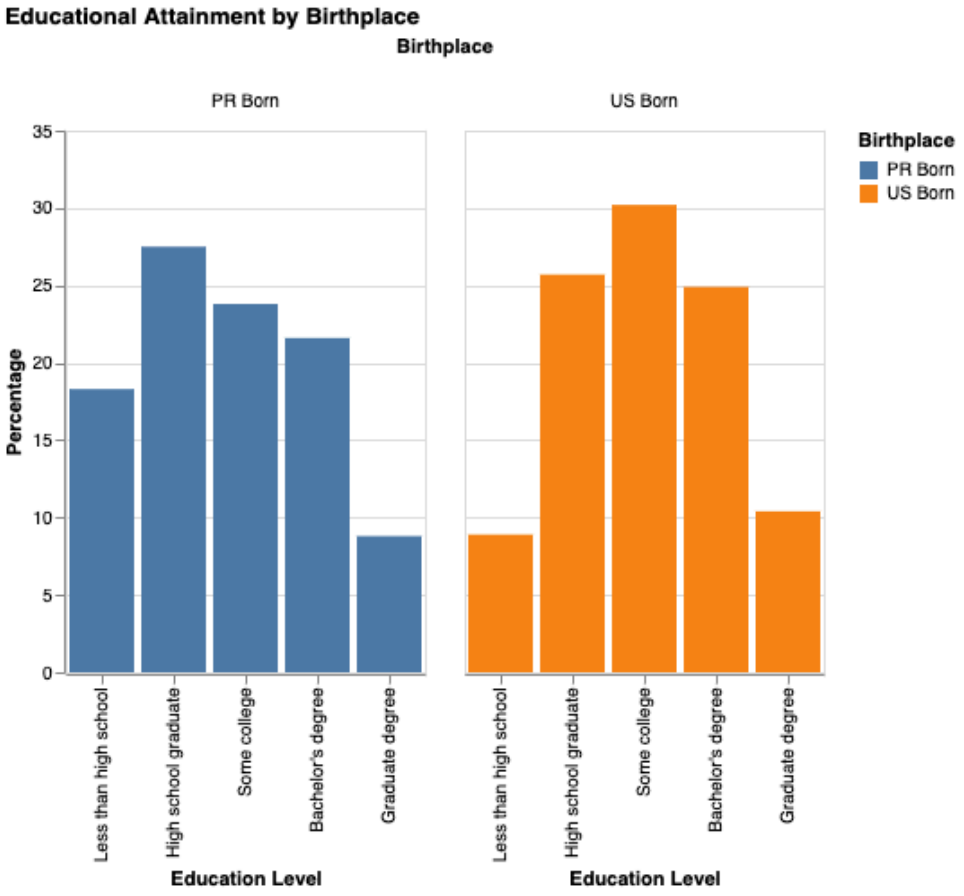
Figure 2: English Language Spoken at Home



This visualization shows the English language proficiency among Puerto Rican born and US mainland born residents of Puerto Rico. Almost all Puerto Ricans do not speak English, and even less of them speak English "very well" when compared to US mainland natives.

Source: United States Census Bureau

Figure 3: Educational Attainment by Birthplace



This graph demonstrates the educational attainment of Puerto Rico born and US born residents of Puerto Rico. The US born population demonstrates a normal curve, while the PR born is skewed more towards the right. It shows that US born residents have higher educational attainment than Puerto Ricans.

Source: United States Census Bureau

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